

HUY THE NGUYEN

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EDUCATION

Bachelor of Arts, Augustana College

Rock Island, IL

Data Science — GPA: 3.5/4.0

Expected Graduation: May 2027

Research Abroad: Aquincum Institute of Technology (Hungary, Jan 2026 – May 2026) — Research Topic: *Machine Learning for Meme Template Identification* — Supervisor: Roland Molontay

Relevant Courses: Machine Learning, Deep Learning, Linear Algebra, Multivariable Calculus, Probability & Statistics, Principles of Computer Systems.

PUBLICATIONS

C = Conference, J = Journal, W = Workshop

[C] **Huy Nguyen**, Donát Ákos Köller, Levente Murgás, József Pintér, Marcell Nagy, Kate Barnes, Roland Molontay. *Meme Template Identification in the Wild: Comparing Methods for Semi-Open-Set Recognition*. Submitted, 2026.

[J] **Huy Nguyen**, Dawoon Jeong. *Anchored Exploration in Semiconductor Product Evolution: A Feature-Level Phylogenetic Network*. Manuscript in preparation in Q1 Journal, 2026.

[W] **Huy Nguyen**, Yue Lin. *Is Your Neighborhood Safe? Place-based Stigma in Large Language Models' Urban Safety Judgments*. Accepted to ACM SIGSPATIAL 2026 Urban AI Workshop.

[C] Xiaofan Zhou, **Huy Nguyen**, Bo Yu, Chenxi Liu, Lu Cheng. *Adaptive Stopping for Multi-Turn LLM Reasoning*. Submitted, 2026.

RESEARCH EXPERIENCE

Research Intern — Supervisor: Dr. Dawoon Jeong

Jun 2026 – Present

The Knowledge Lab, The University of Chicago

Chicago, IL

- Proposed **anchored exploration**, a novel framework by constructing a **two-layer phylogenetic network** of 500,206 semiconductor designs from 1953 to 2024, inferring 493,053 ancestor-descendant relationships and 5,852 capability-transfer events from 563 specifications each.
- Constructed and released a **public dataset** of **1.74M** semiconductor products across **817** standardized attributes, spanning from 1930 to 2024.
- Predicted product obsolescence at **0.72 AUC** and capability origin at **0.74** from phylogenetic network, beating the 0.62 achievable from a design's own specifications.

Visiting Student Researcher — Supervisor: Dr. Roland Molontay

Jan 2026 – May 2026

Budapest University of Technology and Economics

Budapest, Hungary

- Improved meme template classification by proposing **SEED**, a **novel unsupervised method** that outperformed prior unsupervised baselines by **6%** and matched CNN-level performance on the ImgFlip benchmark.
- Analyzed meme popularity over time across **53K+ posts** and **1.5K+ templates**, using **KMeans clustering** on smoothed monthly usage trajectories to identify template lifecycles, peak activity, and long-term usage trends.
- Investigated **global vs. local meme context** by applying BERTopic and sentiment/emotion analysis to 79K+ posts, uncovering how captions reshape template meaning over time.

Research Intern — Supervisor: Dr. Lu Cheng

Aug 2025 – Dec 2025

University of Illinois Chicago

Chicago, IL

- Developed **MiCP**, the first conformal prediction framework with an adaptive rejection mechanism for multi-turn LLM reasoning, by calibrating retrieval and prediction-set thresholds for each question's reasoning steps, providing **formal coverage guarantees** across both retrieval and generated responses.
- Validated the framework across **5 QA benchmarks and 3 LLMs**, including ReAct, while maintaining target coverage and reducing reasoning turns by up to **40.7%** through adaptive early stopping.
- Reduced answer uncertainty through answer clustering and calibrated thresholds, shrinking prediction-set size by up to **22.1%**.

LEADERSHIP & CAMPUS INVOLVEMENTS

Computer Science Grader

Sep 2026 – Present

Augustana College

Rock Island, IL

- Evaluate Python notebooks and written assignments for **CSC-325: Machine Learning**, assessing students' understanding of Bayes' theorem, probabilistic reasoning, classification models, and model evaluation using consistent course rubrics.
- Collaborate with the course professor to provide precise, constructive feedback, identify recurring conceptual and implementation errors, and maintain fair, detail-oriented grading across technical and mathematical coursework.

Technical Lead, Google Developer Group

Sep 2024 – Present

Augustana College

- Led a team of **100+ student organizers** to plan and deliver workshops, technical presentations, and community events focused on computer science, artificial intelligence, and emerging research topics.
- Designed and hosted career-development events such as **“Resume Roast,”** where students rotated through peer-review booths to receive targeted resume feedback for internships, graduate programs, and PhD applications.
- Developed and delivered sessions on **AI-assisted research workflows**, teaching students how to use modern AI tools for conducting research (e.g. literature review, paper comprehension) while promoting responsible and reproducible research practices.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C#, C++.

Developer Tools: Azure, AWS, Git, Tableau, Docker, Microsoft SSMS, NVIDIA CUDA, Apple MPS.

Machine Learning: Scikit-learn, TensorFlow, Keras, PyTorch, Transformers, LangGraph, QLoRA.